

Supervised Machine Learning for Severe Maternal Morbidity Prediction Using Clinician-Curated Obstetric Anesthesia Variables in the nuMoM2b Cohort

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Clinical Question

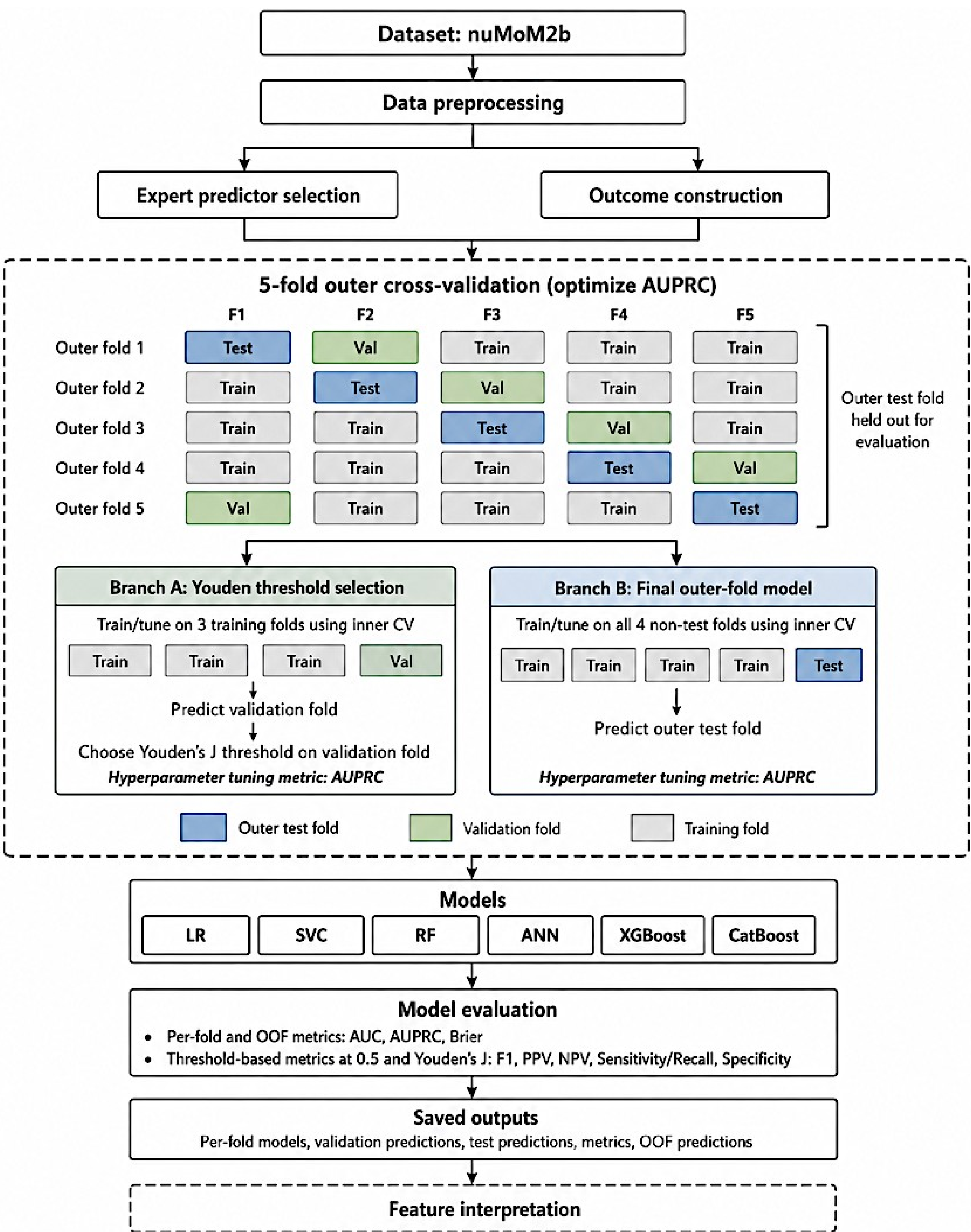
Can 35 clinician-curated obstetric anesthesia variables predict severe maternal morbidity using routinely available clinical data?

9,289
nuMoM2b
Participants

9.46%
Composite SMM
Prevalence

35
Clinician-Curated
Predictors

Nested Validation Design



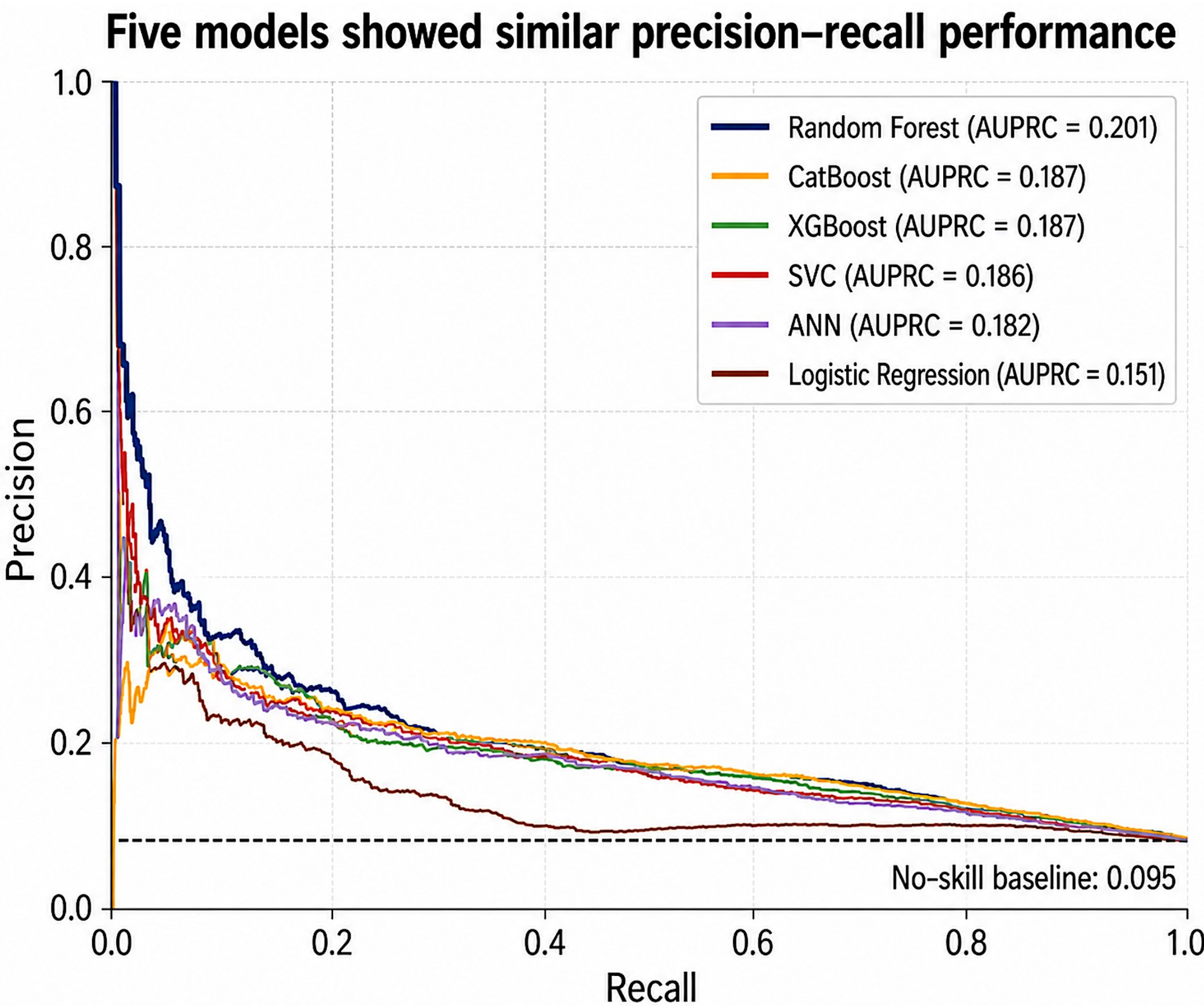
Models: LR, SVC, RF, ANN, XGBoost, and CatBoost.

Primary metric: Pooled OOF AUPRC; prevalence baseline = 0.095.

Leakage control: All preprocessing and SMOTEENN were restricted to training folds. Resampling: SMOTEENN underperformed class-weighted training and was dropped (CatBoost sensitivity at default threshold 0.117 vs. 0.585).

Outcome: Composite SMM based on CDC's 21 indicators; 18 indicators fully represented and 3 indirectly proxied.

Model Performance



CatBoost Had the Best Balanced Operating-Point Performance

Model	Sensitivity	Specificity	F1	MCC
CatBoost	0.618	0.669	0.258	0.175
XGBoost	0.643	0.620	0.244	0.157
Random Forest	0.558	0.692	0.247	0.155
SVC	0.598	0.647	0.240	0.148
ANN	0.569	0.672	0.242	0.148
Logistic Regression	0.511	0.539	0.172	0.029

Metrics calculated from pooled OOF predictions using fold-specific Youden thresholds.

Why CatBoost?

Comparable AUPRC, with the highest F1 and MCC and a stronger sensitivity-specificity balance at fold-specific Youden thresholds.

TAKE-HOME

Leading models showed similar discrimination
Random Forest had the highest observed AUPRC

CatBoost provided the best overall balance
It achieved the highest F1 and MCC at fold-specific Youden thresholds

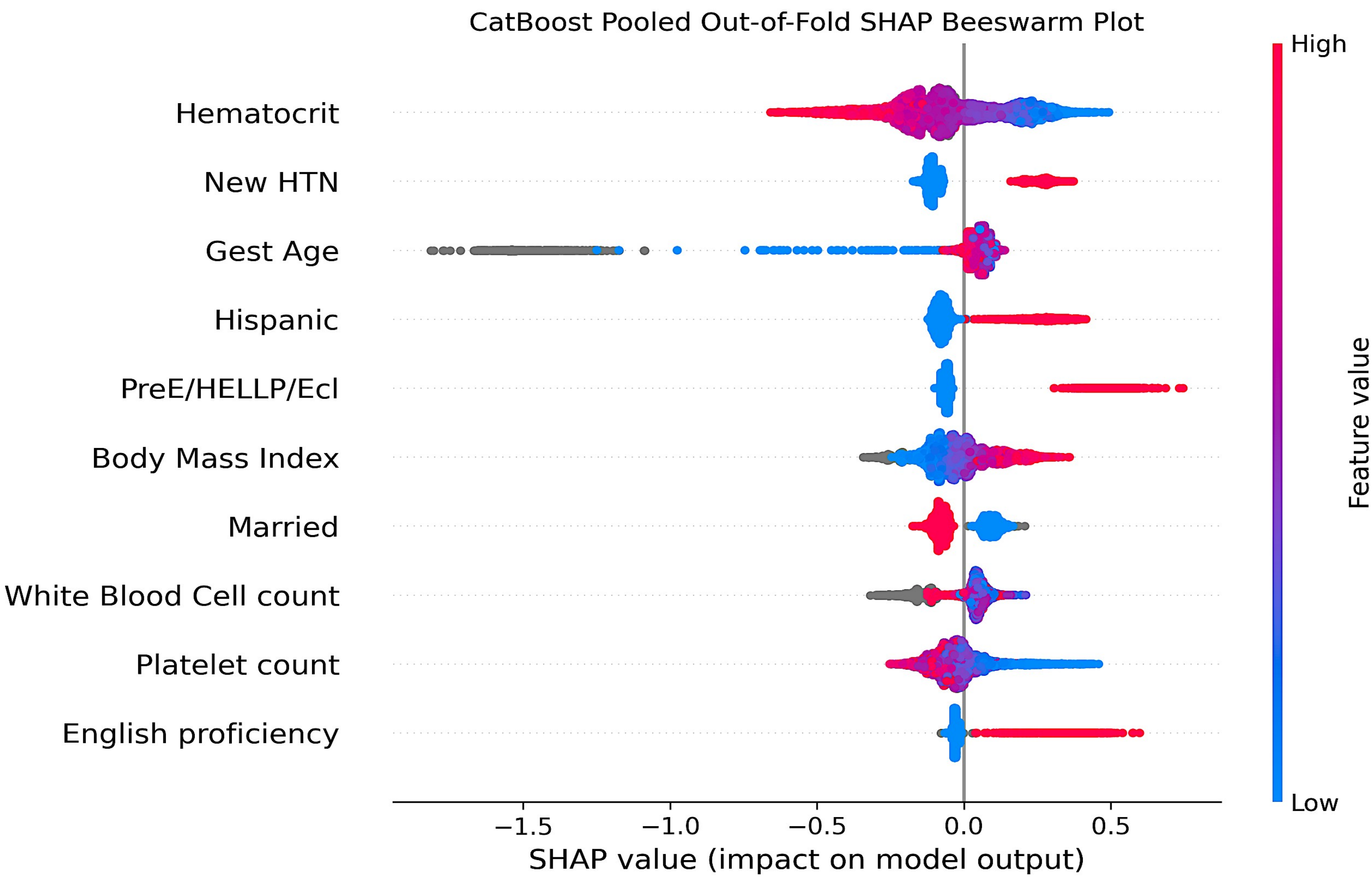
Feature discovery may matter more than
algorithm choice

Overall performance remained modest with the
clinician-curated predictor set

KEY LIMITATIONS

- Internal validation only; external transportability is unknown.
- Survey/chart-derived SMM definitions and three indirect proxies may introduce misclassification.
- Variation in clinical testing and data availability may affect performance and interpretability.
- Modest discrimination, probability recalibration, and subgroup/fairness evaluation must be addressed before clinical deployment.

Model Explainability



At the selected threshold, what would screening look like?

Illustration per 1,000 participants at the observed SMM prevalence (9.46%)

	SMM	No SMM	
Flagged	58 SMM correctly flagged	300 Non-SMM incorrectly flagged	PPV 16.3%
Not flagged	36 SMM cases missed	606 Non-SMM correctly not flagged	NPV 94.4%
	Sensitivity 61.8%	Specificity 66.9%	

Approximate counts derived from observed prevalence and pooled OOF performance; not a separate analysis

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